

ARYAN SARAF

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EDUCATION

University of Southern California

Jan 2025 – Dec 2026

M.S. Computer Science

Los Angeles, CA

- Coursework: Networked Systems for Cloud Computing, Analysis of Algorithms, Applied NLP, Multimedia Systems Design

Shah and Anchor Kutchhi Engineering College

Aug 2019 – May 2023

B.E. Information Technology

Mumbai, India

- Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks

TECHNICAL SKILLS

Languages: Java, Python, GO, C#, Bash

Frameworks: .NET, Unity, PyTorch, Keras, Scikit-Learn

Databases: MySQL, PostgreSQL, MongoDB, Redis, Cassandra, ScyllaDB, Elasticsearch

Cloud & DevOps: GCP, Docker, Kubernetes, Terraform, Ansible, Kafka, Azure DevOps, ELK Stack

Systems & Networking: TCP/IP, BGP, OSPF, DNS, SDN/OpenFlow, Linux, Git

PROFESSIONAL EXPERIENCE

University of Southern California

Jan 2026 – Present

Course Grader – CSCI 526 Advanced Game Consoles

Los Angeles, CA

- Evaluated Unity/C# projects and system designs for 60+ graduate students, using a detailed rubric and providing structured technical feedback that helped students refine their implementations.
- Held weekly office hours to clarify assignment requirements and debug student code, improving average submission quality across the cohort

University of Southern California

May 2025 – Aug 2025

Graduate Research Assistant

Los Angeles, CA

- Designed and shipped CI/CD pipelines (GitHub Actions) for a C#/.NET backend + Unity frontend, cutting manual deployment time by ~40%
- Built dynamic facility upgrade flows with SQLite and Python-driven JSON configs; improved developer onboarding via clean Unity prefabs, reducing integration time by ~40%

JIO Platforms Limited

Oct 2023 – Dec 2024

DevOps Engineer

Mumbai, India

- Architected real-time CDC pipeline from NoSQL datastores to Elasticsearch, enabling live data analytics for enterprise clients
- Automated GCP infrastructure provisioning with Terraform, reducing deployment time by ~80% and standardizing environments across Dev, QA, Pre-Prod, and Prod
- Designed and deployed CI/CD pipelines in Azure DevOps + Google Artifact Registry, cutting platform latency by 30% through zero-downtime Kubernetes rollouts
- Audited and decommissioned idle servers, saving \$7,000+ in monthly cloud costs; conducted vulnerability and malware assessments across 100+ production servers
- Managed and tuned MySQL, PostgreSQL, Cassandra, ScyllaDB, MongoDB, Redis, and Elasticsearch clusters, maintaining high availability across all pipeline stages

ACADEMIC PROJECTS

Latency-Aware Load Balancer

GO, Mininet, Python, OpenFlow

- Designed and built a latency-aware load balancer in Go on a fat-tree datacenter topology (20+ nodes) simulated in Mininet with an SDN/OpenFlow control plane
- Implemented real-time probe-based latency sampling; dynamic routing reduced average response time by **38%** vs. static round-robin under bursty load
- Benchmarked round-robin, least-latency, and straggler-mitigation strategies; least-latency achieved **2.1×** throughput improvement at the 95th percentile tail latency

Adaptive Query Reformulation (AQR-RAG)

Python, DistilRoBERTa, RAG

- Built a dynamic query-rewriting pipeline classifying weak vs. strong queries using fine-tuned DistilRoBERTa (82.4% accuracy, 0.90 PR-AUC)
- Integrated Google Trends API for real-time context injection, improving retrieval precision on ambiguous queries with trend-aligned terminology